

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15B.3

NUCLEOPHILIC ADDITION REACTIONS

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: ORGANIC CHEMISTRY, CARBONYLS AND CHIRALITY
SUB-TOPIC:	15B.3: Nucleophilic Addition Reactions
TOTAL QUESTIONS:	3 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	9 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15B.3

Q1. 1. (a) Draw the mechanism for the reaction of ethanal with HCN in the presence of KCN catalyst to form 2-hydroxypropanenitrile.

[4 Marks]**WCH14/01/Jan22/Q7(a)**

Q2. (b) Describe the observation and result when propanone reacts with 2,4-dinitrophenylhydrazine (2,4-DNPH).

[2 Marks]**WCH14/01/Jan22/Q7(b)**

Q3. (c) Describe the iodoform (tri-iodomethane) test for methyl ketones.

[3 Marks]**WCH14/01/Jan22/Q7(c)**

SECTION B: WORKED MARK SCHEME — 15B.3

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Draw the mechanism for the reaction of ethanal with HCN in the presence of KCN catalyst to form 2-hydroxypropanenitrile. 1. Nucleophilic attack by CN ⁻ lone pair on delta+ carbonyl carbon. 2. Curly arrow from C=O bond onto oxygen atom. 3. Formation of tetrahedral intermediate [CH ₃ CH(O ⁻)CN]. 4. Protonation of O ⁻ by H ⁺ / HCN to form -OH group.	[4]
Q2	Q2: (b) Describe the observation and result when propanone reacts with 2,4-dinitrophenylhydrazine (2,4-DNPH). 1. Orange / yellow precipitate forms. 2. Confirms presence of carbonyl group (C=O aldehyde or ketone).	[2]
Q3	Q3: (c) Describe the iodoform (tri-iodomethane) test for methyl ketones. 1. Reagent: I ₂ and NaOH (aq). 2. Observation: Yellow precipitate of CHI ₃ with antiseptic smell. 3. Confirms presence of CH ₃ C=O or CH ₃ CH(OH) group.	[3]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 15B.3**FAQ #1: Why is KCN added in addition to HCN in nucleophilic addition?****Category: Reaction Conditions****COMMON EXAMINER TRAP / MISTAKE:**

Thinking HCN alone is reactive enough.

MODEL ANSWER / EXAMINER EXPECTATION:

HCN is a weak acid and dissociates poorly. KCN provides a high concentration of CN^- nucleophiles to initiate the reaction.